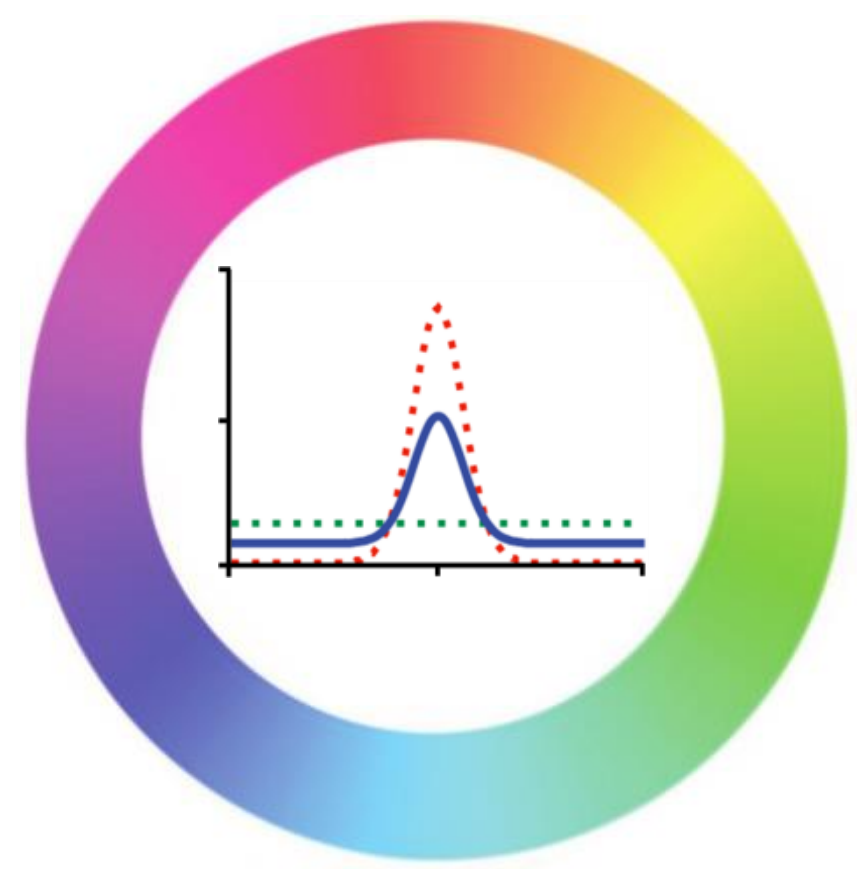




Visual Working Memory for Configural information

Inik Kim, Weiwei Zhang

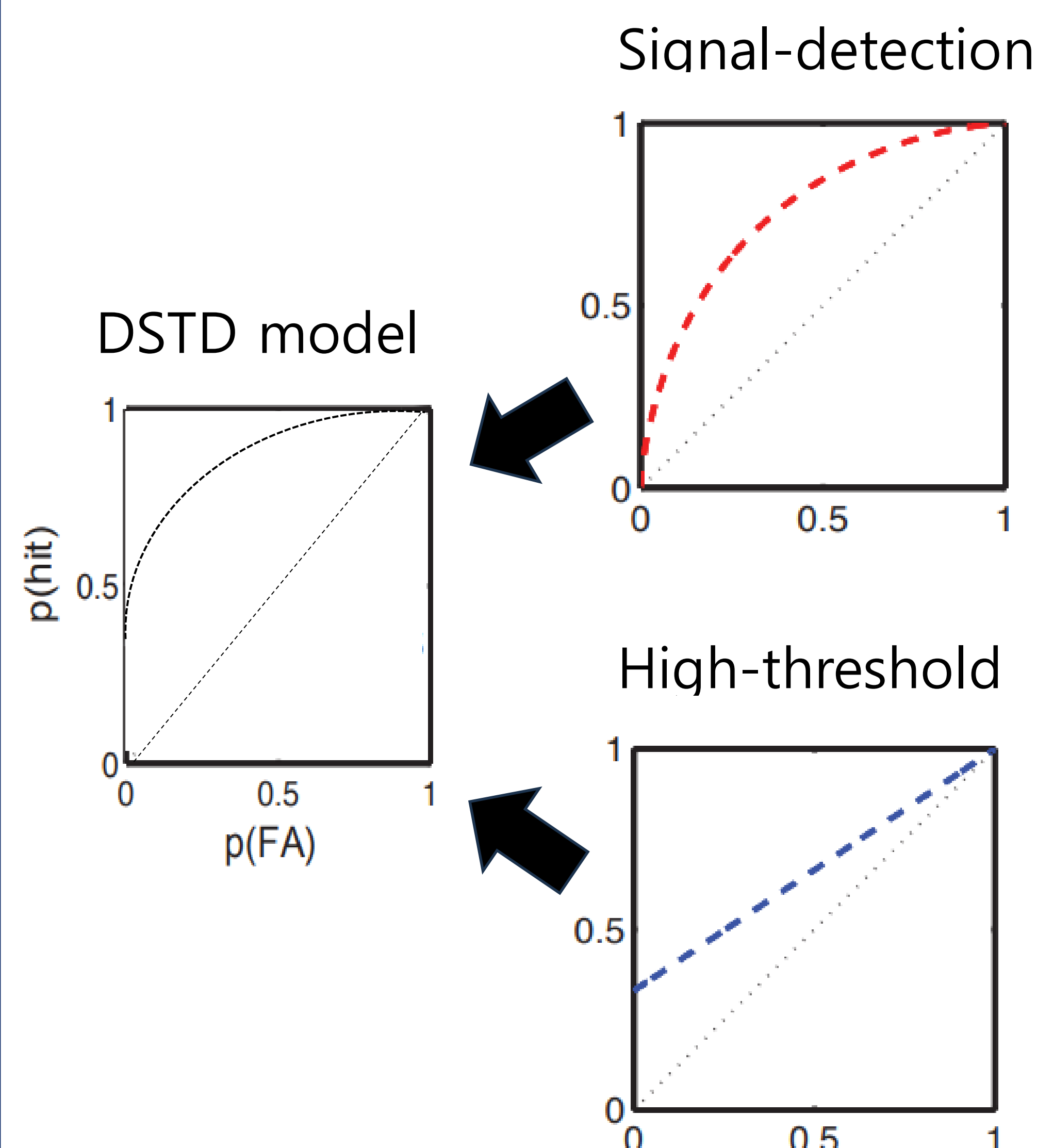
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INTRODUCTION

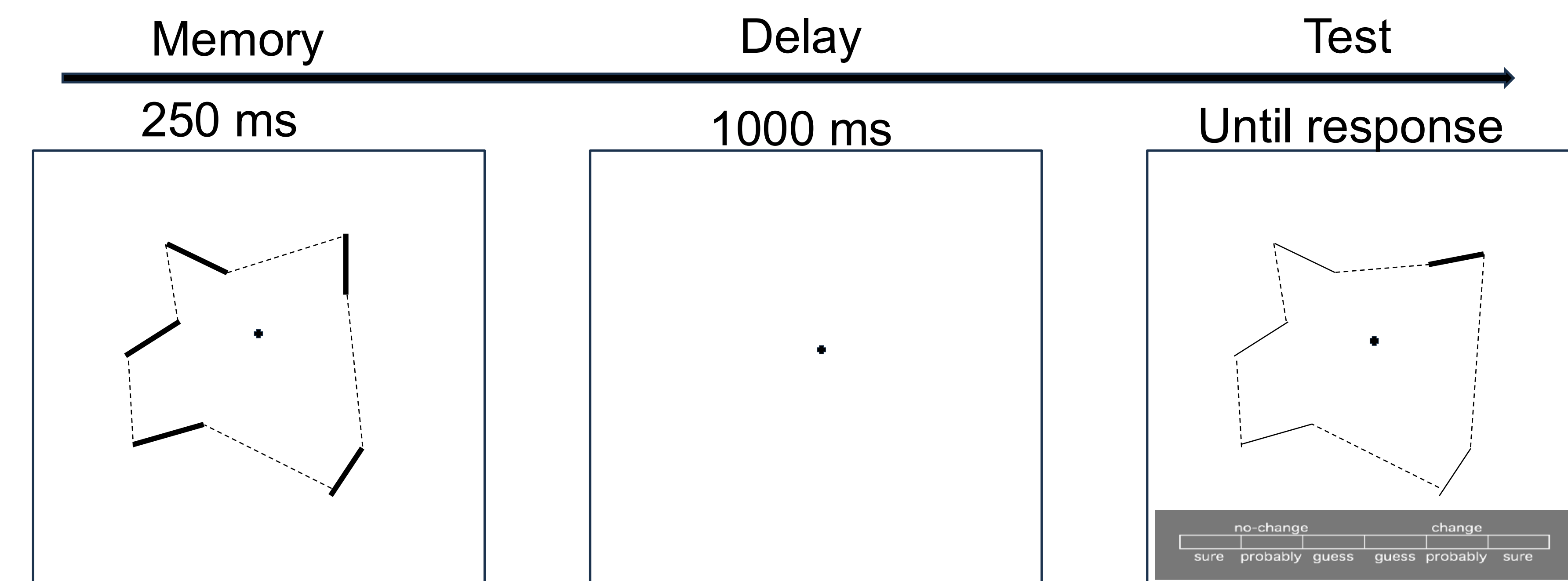
- Configural cues can enhance global VWM representations (e.g., the overall shape of the orientation bars) with little effect on local item representations (Xie & Zhang, 2017).
- Xie & Zhang (2017) model (see below) captures global and local representations in VWM by decomposing ROC data into signal detection theory and high-threshold components, respectively.
- The present study introduced competition between configural cue and global VWM representations by manipulating the relevancy of configural cues.
- Irrelevant cue is expected to reduce global VWM representations.

DTSD (dual trace signal detection)



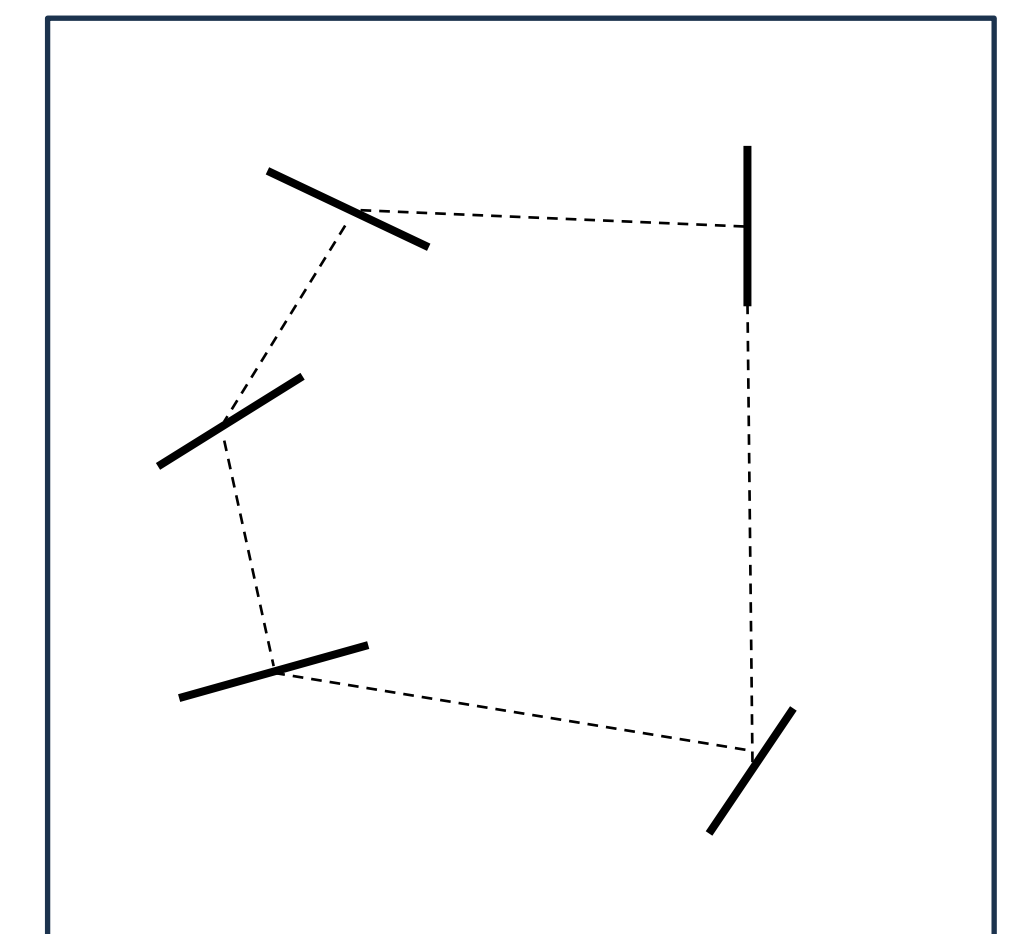
EXPERIMENT

Task procedure



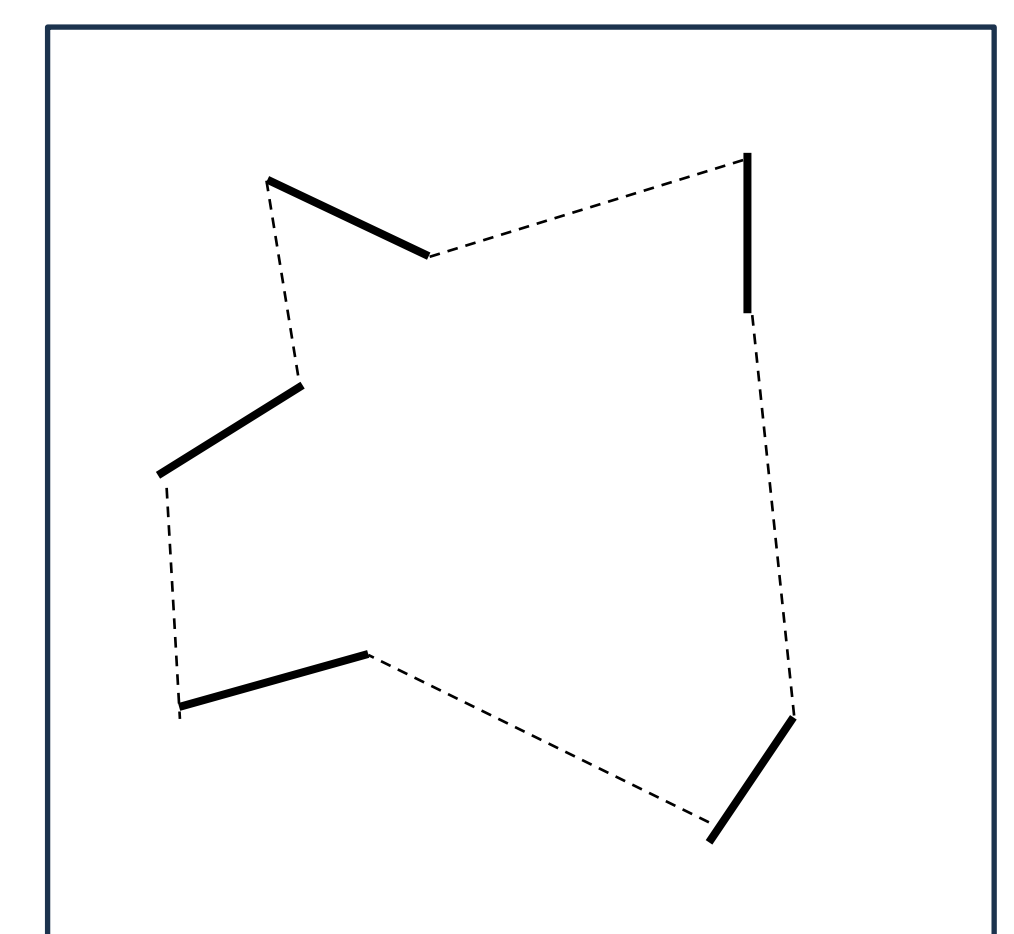
Irrelevant configuration

- Configural cues (dash lines) are connected to the center of target stimuli.
- Configuration or overall shape is independent of local orientations.



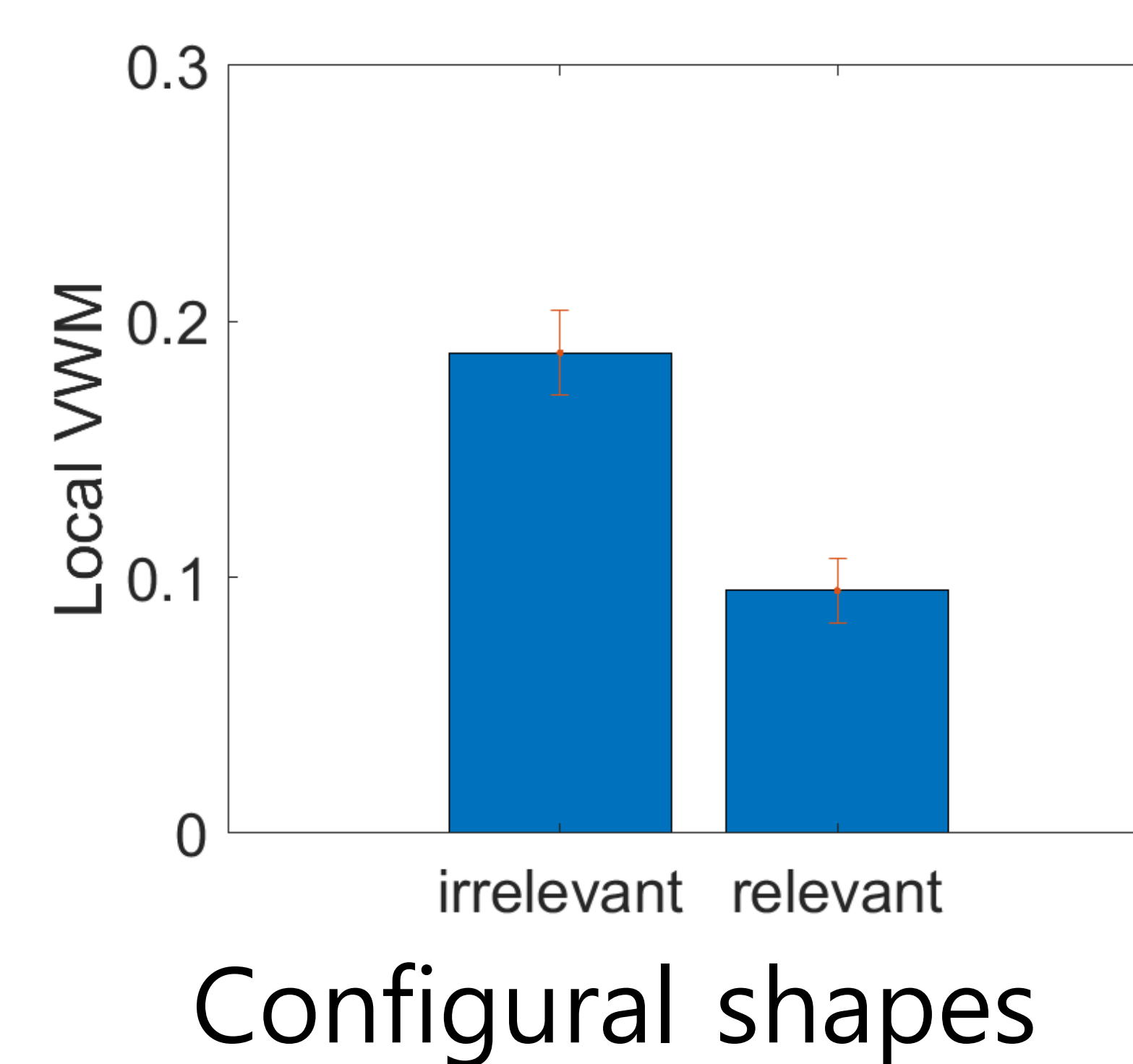
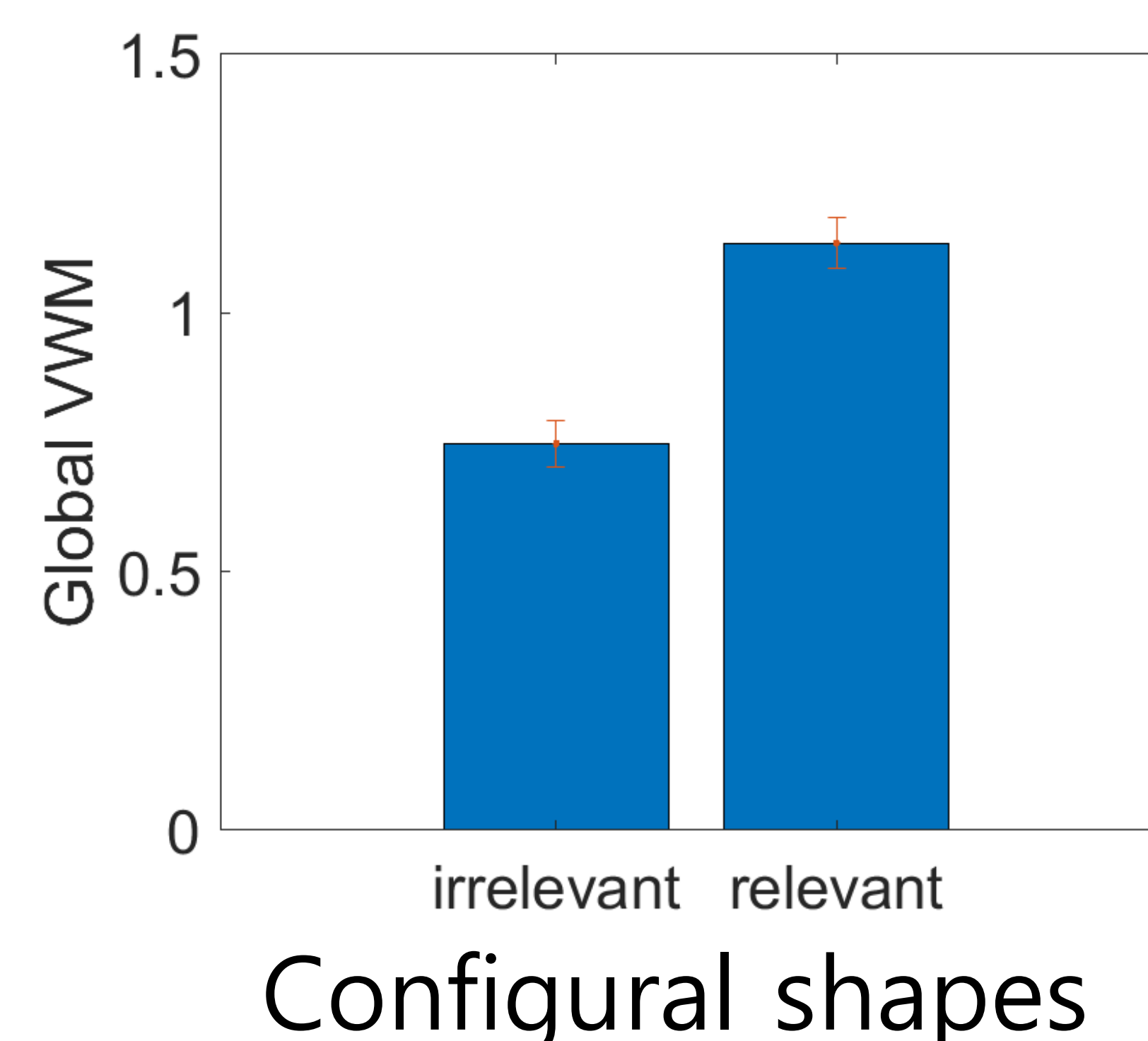
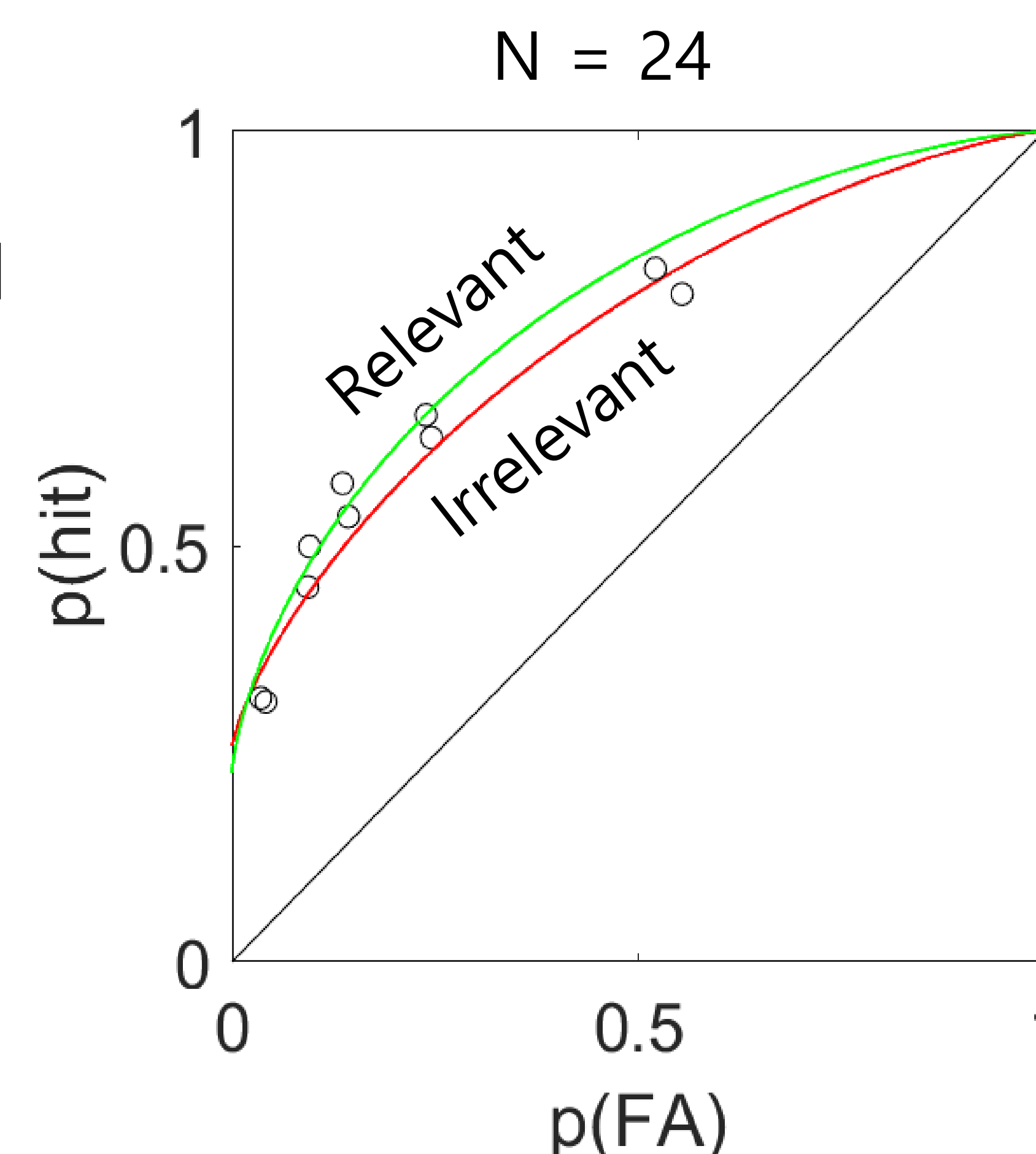
Relevant configuration

- Configural cues are connected to each end of target stimuli.
- Configuration shape is relevant to local orientations.



Results

- The DTSD model was applied to raw ROC data and plotted in the graph at right.
- Global VWM representation was worse in the irrelevant configuration condition whereas local VWM representation was improved.



* <.05, ** <.01, *** <.001

CONCLUSIONS & DISCUSSION

- Configuration effect in VWM was re-examined by manipulating the relevancy of configuration.
- Trade off global WM representations for local VWM representations by irrelevant configural cues ← the competition between irrelevant cue and global VWM representations
- Taken together, we have identified some boundary conditions for the impact of configural cue on VWM